

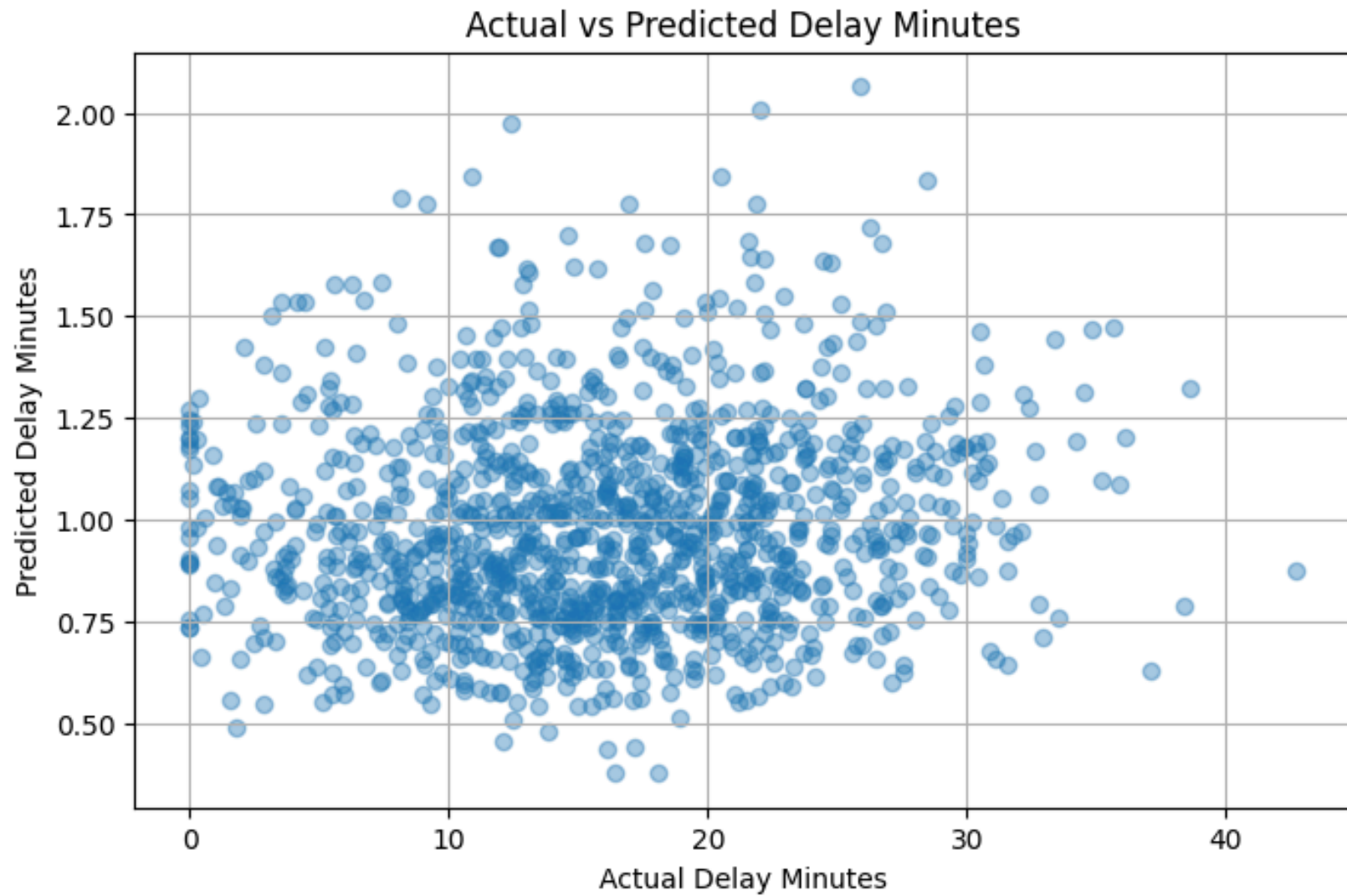
Transportation Telemetry Predictive Timing Platform

Synthetic train telemetry, lakehouse pipeline, and neural network delay prediction

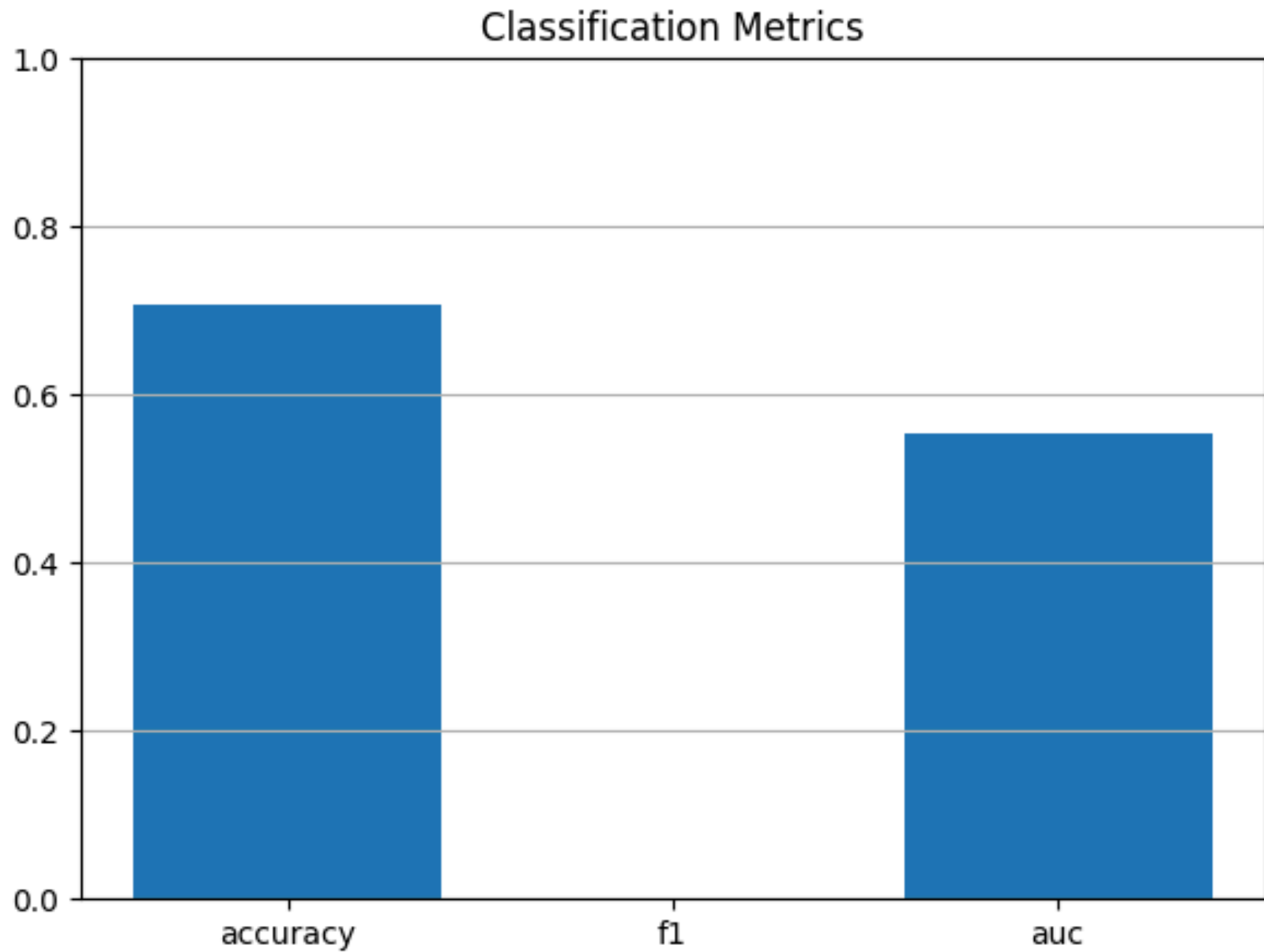
Project question

Can a real-time transportation telemetry platform predict train delay timing and support predictive maintenance decision-making using sensor, location, and operational event data?

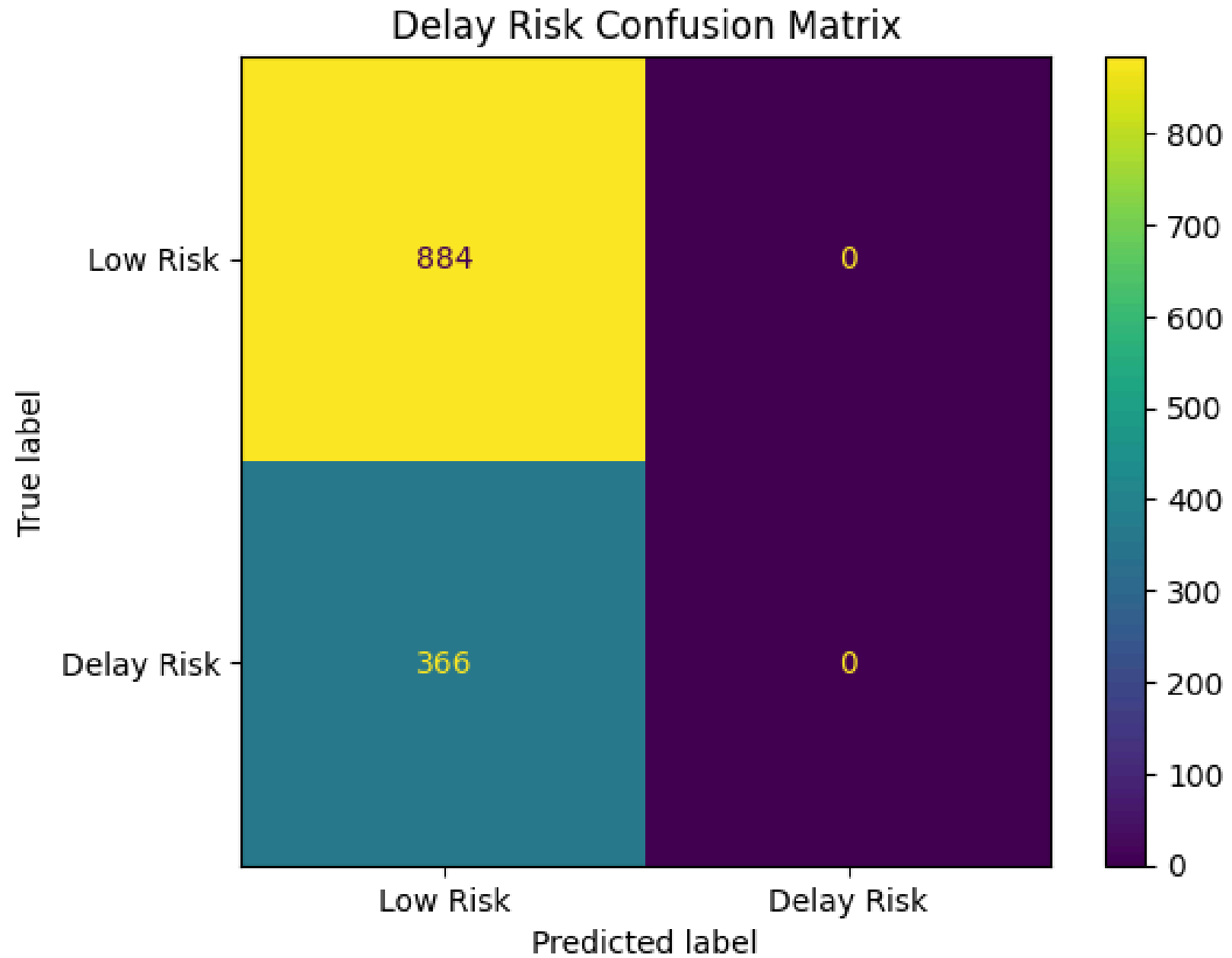
Overview Metrics **Figures** Architecture



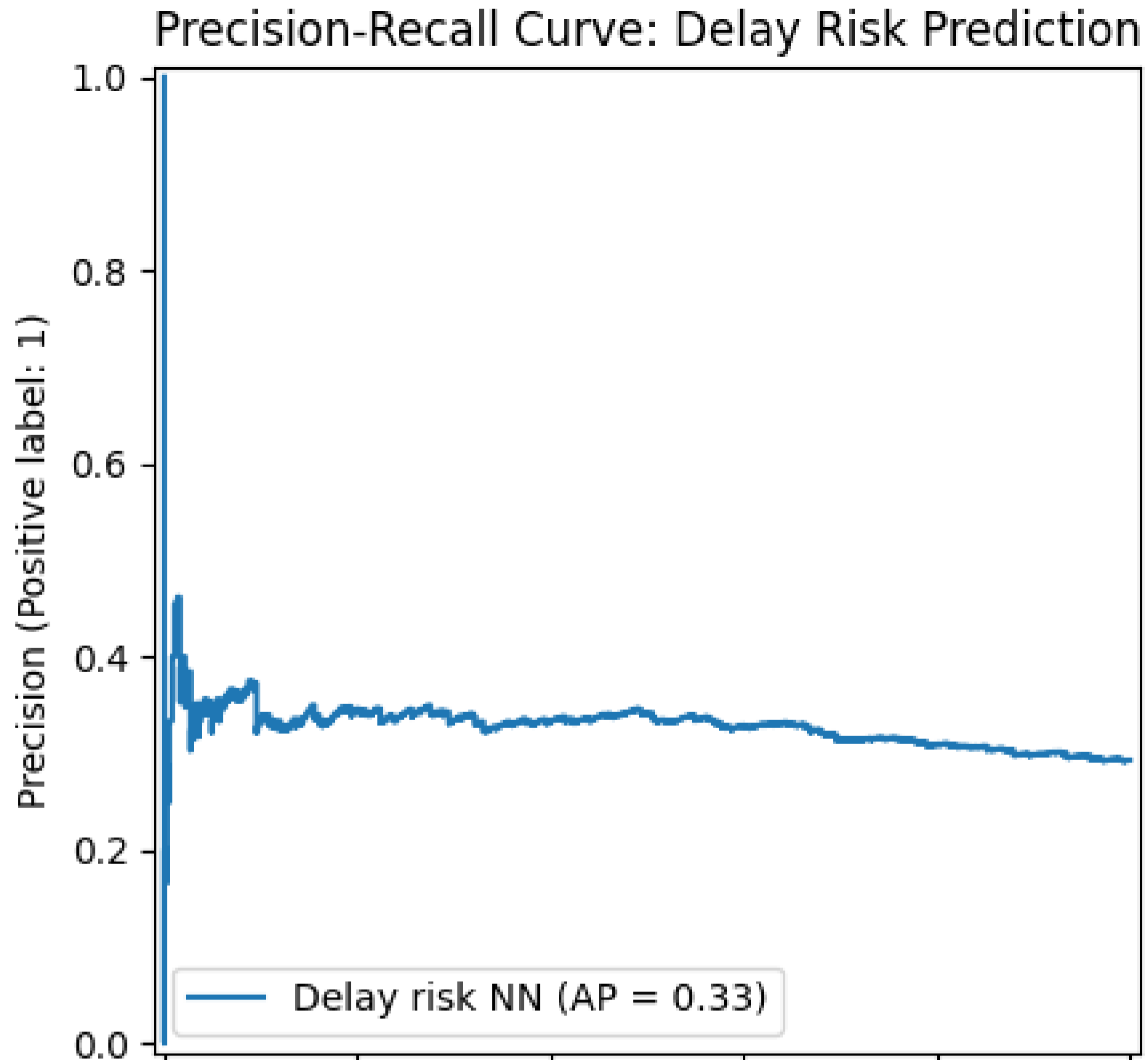
actual_vs_predicted_delay_minutes.png

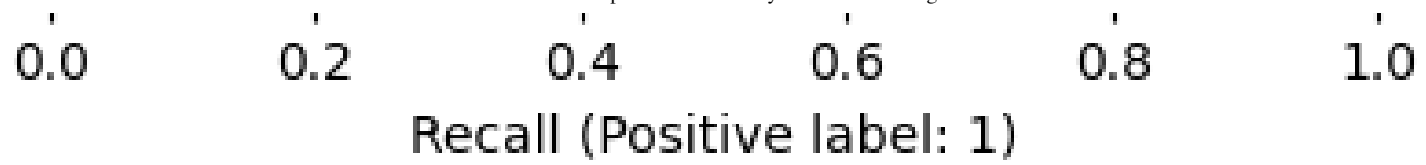


classification_metrics_bar_chart.png



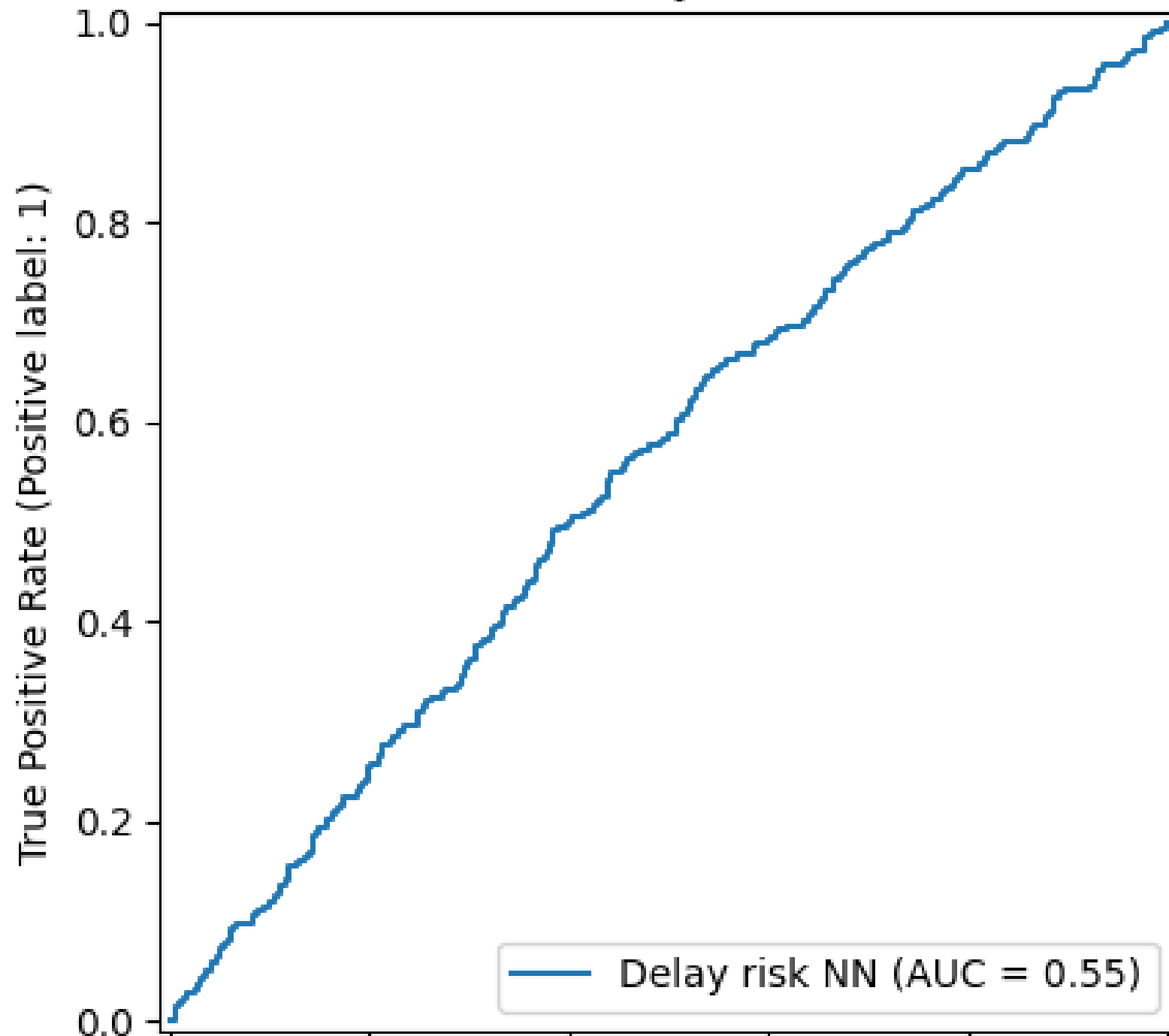
confusion_matrix_delay_risk.png





precision_recall_delay_risk.png

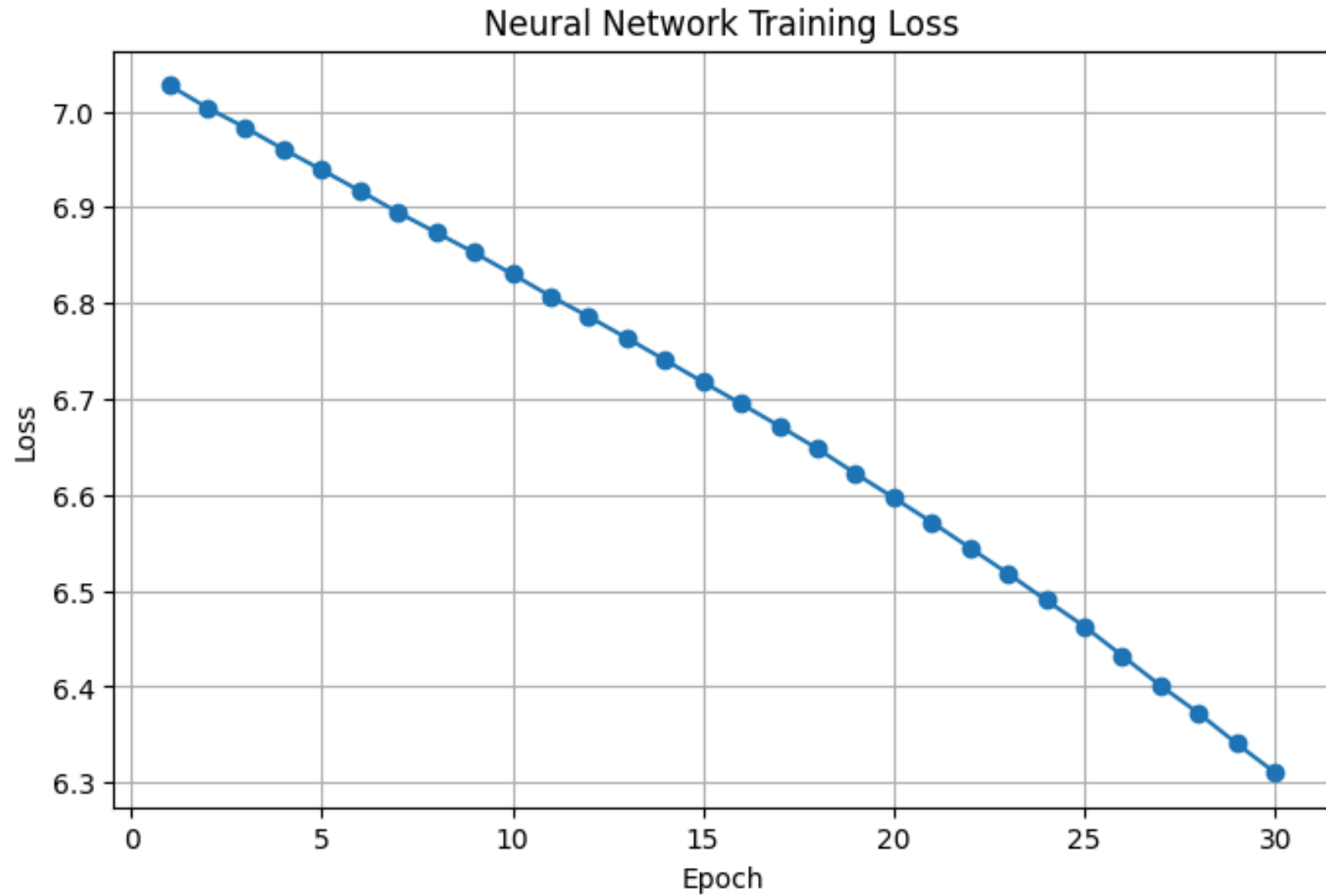
ROC Curve: Delay Risk Prediction



0.0 0.2 0.4 0.6 0.8 1.0

False Positive Rate (Positive label: 1)

roc_curve_delay_risk.png



training_loss_curve.png

